PROPERTIES OF DUAL PROBLEMS

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Property I

- At any simplex iteration of the primal or the dual,

Objective coefficient of variable *j* in one problem= Left-hand side minus right-hand side of constraint *j* in the other problem

- Also, remember, dual of the dual is primal.
- It will be computationally efficient to solve a problem with less number of constraints.

Property II

- For any pair of feasible primal and dual solutions,

(Objective value in the maximization problem) ≤(Objective value in the minimization problem)

Equality holds at the optimum.

Minimize $z = 40y_1 + 24y_2 + 10y_3$

s.t.

$$y_1 + 3y_2 + y_3 \geq 2,$$
$$3y_1 + y_2 + y_3 \geq 5,$$
$$y_1 \geq 0, y_2 \geq 0, y_3 \geq 0.$$

By solving the dual form using simplex method, find out the minimum value of z and corresponding y_1, y_2 and y_3. Which are the active constraints of primal problem? Find out the unit worth of active constraints.

Dual form:

$$\text{Maximize } w = 2x_1 + 5x_2$$

s.t.

$$x_1 + 3x_2 \le 40,$$
$$3x_1 + x_2 \le 24,$$
$$x_1 + x_2 \le 10$$
$$x_1 \ge 0, x_2 \ge 0.$$

We introduce slack variables to convert the dual form into standard form.

$$\text{Maximize } w = 2x_1 + 5x_2 + 0x_3 + 0x_4 + 0x_5$$

s.t.

$$x_1 + 3x_2 + x_3 = 40,$$
$$3x_1 + x_2 + x_4 = 24,$$
$$x_1 + x_2 + x_5 = 10,$$
$$x_1, x_2, x_3, x_4, x_5 \ge 0.$$

Basic	x_1	x_2	x_3	x_4	x_5	Solution
w	-2	-5	0	0	0	0
x_3	1	3	1	0	0	40
x_4	3	1	0	1	0	24
x_5	1	1	0	0	1	10

Basic	x_1	x_2	x_3	x_4	x_5	Solution
w	3	0	0	0	5	50
x_3	-2	0	1	0	-3	10
x_4	2	0	0	1	-1	14
x_2	1	1	0	0	1	10

The optimal solution is $x_1 = 0$ and $x_2 = 10$. Maximum $w = 5 \times 10 = 50$.

In the dual problem, the optimal values of the slack/surplus variables in the objective function (w-row) correspond to the values of the primal variables i.e., $y_1 = 0$, $y_2 = 0$, and $y_3 = 5$.

How? By Property I.

Corresponding to x_3, we have $y_1 \geq 0$. Hence, $0 = y_1 - 0$

Corresponding to x_4, we have $y_2 \geq 0$. Hence, $0 = y_2 - 0$

Corresponding to x_5, we have $y_3 \geq 0$. Hence, $5 = y_3 - 0$

The primal constraint $y_1 + 3y_2 + y_3 \geq 2$ (corresponding to dual variable x_1) is inactive (since $x_1 = 0$).

The primal constraint $3y_1 + y_2 + y_3 \geq 5$ (corresponding to dual variable x_2) is active (since $x_2 = 10$).

The unit worth of active constraint $3y_1 + y_2 + y_3 \geq 5$ is $x_2 = 10$ (corresponding dual variable value).

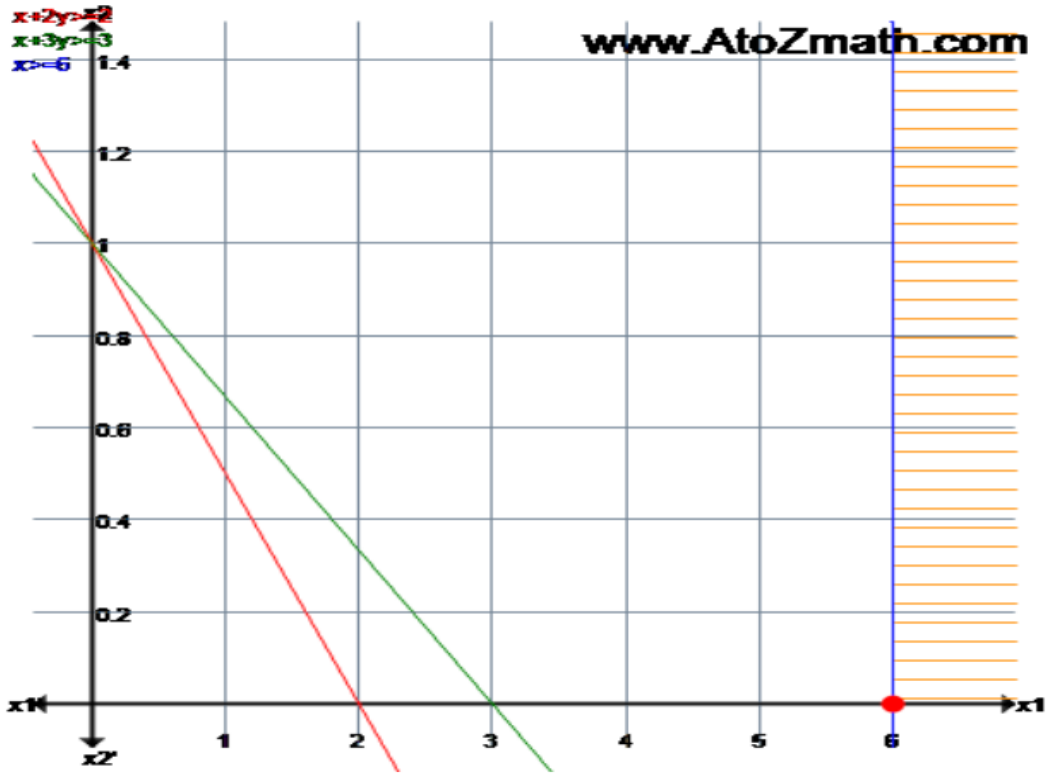
Maximize $z = 2x_1 + 3x_2 + 6x_3,$

subject to $x_1 + x_2 + x_3 = 1;\ 2x_1 + 3x_2 = 1;\ x_1 \geq 0, x_2 \geq 0, x_3 \geq 0.$

DUAL :

Minimize $w = y_1 + y_2;$

subject to $y_1 + 2y_2 \geq 2; y_1 + 3y_2 \geq 3; y_1 \geq 6; y_1, y_2 \geq 0.$


x+2y<=2
x+3y<=3
x<=6
www.AtoZmath.com
1.4
1.2
1
0.8
0.6
0.4
0.2
x1
x2
1
2
3
4
5
6
x1

The value of the objective function at each of these extreme points is as follows:

Extreme Point Coordinates (x_1, x_2)	Lines through Extreme Point	Objective function value $Z = x_1 + x_2$
$A(6, 0)$	$3 \rightarrow x_1 \geq 6$ $5 \rightarrow x_2 \geq 0$	$1(6) + 1(0) = 6$

The miniimum value of the objective function $Z = 6$ occurs at the extreme point $(6, 0)$.

Hence, the optimal solution to the given LP problem is : $x_1 = 6, x_2 = 0$ and min $Z = 6$.

Unit worth of resource 1 = 6 and unit worth of resource 2 = 0.

Proof of Property II

Primal

Minimize $\mathbf{c}^T \mathbf{x}$

Subject to $\mathbf{Ax}=\mathbf{b}$

$\mathbf{x} \geq 0$

Dual

Maximize $\mathbf{y}^T \mathbf{b}$

S. t. $\mathbf{y}^T\mathbf{A} \leq \mathbf{c}^T$

Lemma

- If **x** any **y** are feasible for primal and dual respectively, then $\mathbf{c}^T \mathbf{x} \geq \mathbf{y}^T \mathbf{b}$.
- Proof: We have
- $\mathbf{y}^T \mathbf{b} = \mathbf{y}^T \mathbf{A} \mathbf{x} \leq \mathbf{c}^T$
- This is true because $\mathbf{x} \geq \mathbf{0}$

Duality theorem

- If either of primal and dual has a finite optimal solution, so does the other, and the corresponding values of the objective functions are equal. If either problem has an unbounded objective, the other problem has no feasible solution.

An example

- Maximize $z=2x+3y$

Subject to $x-4y \leq 10$

$x,y \geq 0.$

You can take a very big value of y without violating constraint. Hence, z will become unbounded.

Let us make a dual problem. First write in standard form.

Maximize $z=2x+3y+0S$

Subject to $x-4y+S=10$, (where S is a slack variable. Notice that right hand side is positive.)

$x,y,S \geq 0.$

Dual of Standard Problem

- Minimize $w=10\ t$
- Subject $t \geq 2;\ -4t \geq 3;\ t \geq 0$.
- However, $t \geq 2$ and $-4t \geq 3$ cannot be satisfied together. Hence, solution is infeasible.

- Note, $-4t \geq 3 \Rightarrow t \leq (-3/4)$.